451 Financial Engineering

Programming Assignment 1

AAPL Return Prediction

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Goal

This project asks whether machine learning can be used to predict the direction of returns for Apple Inc. common stock (ticker AAPL). Specifically, we predict whether the five-day forward log return will be positive or negative, using a combination of own-price features, volume signals, technical indicators, and cross-asset features drawn from related ETFs, sector indices, and macroeconomic series. We evaluate four imodel builds, each motivated by the limitations of the previous one, and assess performance on a held-out test set comprising the most recent 15% of the data.

Methodology

Exploratory Data Analysis

The training set covers AAPL daily data from January 2010 through October 2023, with the hold-out spanning November 2023 through April 2026. Figure 1 summarizes the two most important properties of the data: the distribution of daily log returns (left) and the path of the adjusted close price (right). The log return distribution is centered close to zero, roughly symmetric, with the heavy tails characteristic of equity returns. The price path shows a roughly thirty-fold increase over the training window, from approximately six dollars in 2010 to nearly two hundred by late 2023. This dramatic non-stationarity in the price level is the single most important fact about the dataset for feature construction purposes: any model relying on raw price as an input will see a hold-out distribution that sits almost entirely outside its training range.

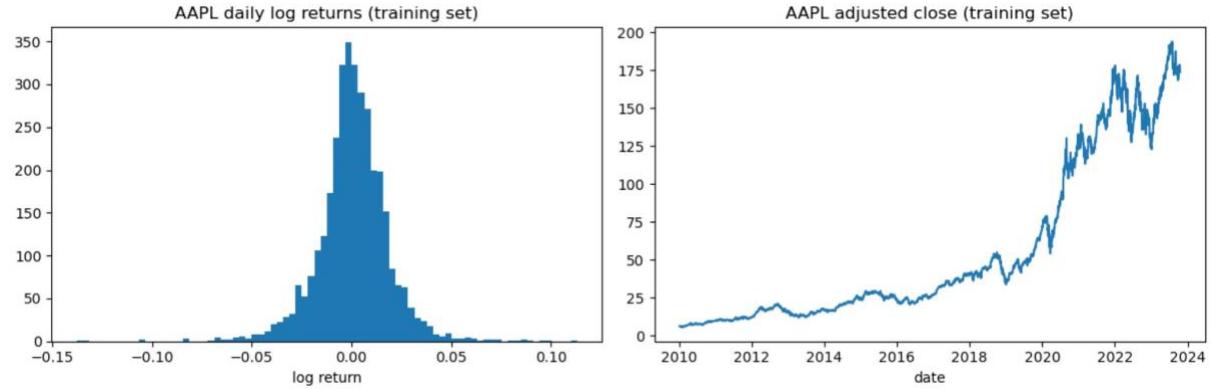

Figure 1. *AAPL daily log returns (left) and adjusted close price (right) over the training window. The log return distribution is approximately stationary; the price level is not.*

Target

We predict the five-day forward log return of AAPL rather than the next-day direction. This choice is motivated by three related observations. First, daily equity returns contain a high proportion of noise relative to any underlying predictable signal. Summing five daily returns partially averages out that noise while preserving persistent directional drift, which improves the signal-to-noise ratio of the target. Second, a five-day horizon corresponds more closely to the rebalancing frequency of swing-trading strategies where machine-learning-driven approaches are actually deployed in practice, so a successful model at this horizon is more economically meaningful than a daily-frequency model. Third, our early iterations at a one-day classification horizon produced hold-out accuracy near 50% regardless of feature configuration, which motivated a direct test of whether a longer horizon would expose signal that daily noise was obscuring.

Starting in Model 3 we predict the binary direction of the five-day return. In Model 4 we switch to regressing the continuous five-day return directly and threshold the prediction to recover a directional call. Regression preserves magnitude and ranking information that binary classification discards and allows evaluation via the information coefficient, the Spearman rank correlation between predicted and realized returns, which is the standard metric in return-forecasting research.

Train and Hold-Out Discipline

Across all four model builds we split the data chronologically into a training set comprising the earliest 85% of observations and a hold-out test set comprising the most recent 15%, with a ten-row gap between them to prevent any rolling-window feature from peeking across the boundary. All feature scaling, hyperparameter search, and cross-validation are performed on the training set only. The hold-out is evaluated exactly once at the end of each model build. Cross-validation within the training set uses a time-series split with five folds and a ten-row gap between train and validation folds, which respects the temporal ordering of the data and prevents lookahead leakage in the CV folds themselves.

Feature Selection

Features fall into four categories: price-based, return-based, technical indicators, and cross-asset. The price-based group was evaluated in our baseline model and dropped thereafter because of distribution-shift issues described in the Model Building section. Rather than re-introducing price-level information in raw form, we captured the same underlying behavioral intuition through stationary transformations such as the percentage distance to the 52-week high, which is already included in the final feature set. The final model uses 29 features drawn from the remaining three categories.

Price-Based Features (dropped after Model 1)

- **CloseLag1, CloseLag2, CloseLag3 —** Lagged closing prices, one to three days back
- **HML —** Daily high minus daily low, in dollars
- **HMLLag1, HMLLag2, HMLLag3 —** Lagged dollar HML
- **OMC —** Daily open minus close, in dollars
- **OMCLag1, OMCLag2, OMCLag3 —** Lagged dollar OMC
- **VolumeLag1, VolumeLag2, VolumeLag3 —** Lagged raw trading volume
- **CloseEMA2, CloseEMA4, CloseEMA8 —** Exponential moving averages of lagged close, with half-lives 1, 2, and 4 days

Return-Based Features

- **RetLag1, RetLag2, RetLag3 —** Lagged daily log returns, one to three days back
- **HMLpctLag1, HMLpctLag2, HMLpctLag3 —** Daily high minus low, divided by close, lagged one to three days
- **OMCpctLag1, OMCpctLag2, OMCpctLag3 —** Daily open minus close, divided by close, lagged one to three days
- **VolRatioLag1, VolRatioLag2, VolRatioLag3 —** Lagged volume divided by trailing 20-day mean volume
- **RetEMA2, RetEMA4, RetEMA8 —** Exponential moving averages of lagged returns, with half-lives 1, 2, and 4 days
- **RetVol20 —** 20-day rolling standard deviation of lagged returns

Technical Indicator Features

- **RSI_14 —** 14-day Relative Strength Index, computed on lagged returns. Measures whether recent gains outpace recent losses.
- **MACD —** Difference between the 12-day and 26-day EMAs of lagged close, normalized by lagged close
- **MACD_Signal —** 9-day EMA of MACD
- **MACD_Hist —** MACD minus MACD_Signal, capturing momentum shifts
- **DistTo52wHigh —** Percentage distance from lagged close to its 252-day rolling maximum

Cross-Asset Features

- **SPY_RetLag1 —** Lagged log return of the S&P 500 ETF
- **QQQ_RetLag1 —** Lagged log return of the Nasdaq 100 ETF
- **XLK_RetLag1 —** Lagged log return of the technology sector ETF
- **SOXX_RetLag1 —** Lagged log return of the semiconductor sector ETF

- **TSM_RetLag1 —** Lagged log return of Taiwan Semiconductor Manufacturing, AAPL's primary chip supplier
- **VIX_LvlLag1 —** Lagged level of the CBOE Volatility Index, a market-wide risk indicator
- **VIX_ChgLag1 —** Lagged one-day change in VIX
- **TNX_ChgLag1 —** Lagged one-day change in the 10-year Treasury yield

Model Building Overview

Model 1 — Baseline

Our baseline model mirrored the WTI reference pipeline: an XGBoost classifier predicting the direction of next-day AAPL returns using price-level lags, dollar-denominated HML and OMC, raw volume lags, and exponential moving averages of the closing price. The in-sample AUC came in at 0.92 but the hold-out AUC dropped to 0.48, a gap indicative of severe overfitting. The cause was straightforward once we examined the data: the non-stationarity visible in Figure 1 means that when the model went to predict on the hold-out period, the price values it was seeing sat well outside the range it had trained on. Tree-based models can split on thresholds they learned during training but cannot meaningfully extrapolate past them. We rejected this feature formulation as a viable model and moved to stationary features in Model 2.

Model 2 — Stationary Features and Initial Cross-Asset Set

Model 2 replaced every dollar-denominated feature with a stationary transformation. Lagged prices became lagged log returns. Dollar HML became HML divided by close. Raw volume became the ratio of yesterday's volume to its 20-day average. EMAs were computed on lagged returns rather than on lagged prices. We also introduced an initial set of cross-asset features: lagged SPY and QQQ returns, VIX level and change, and the change in the 10-year Treasury yield. The in-sample and hold-out AUCs converged to approximately 0.50, confirming that the overfitting problem had been eliminated. However, the converged AUC sat at the random-guessing baseline, so while the methodology was now sound, the model lacked any real predictive signal.

Model 3 — Expanded Cross-Asset Set and Five-Day Horizon

Model 3 made two substantive additions. We expanded the cross-asset set to include SOXX (semiconductor sector ETF), TSM (Taiwan Semiconductor, AAPL's largest chip supplier), XLK (technology sector ETF), and the Nikkei 225 (Japanese market index, intended as a proxy for overnight Asian sentiment since Japan's market closes before the US open). We also introduced the five-day directional target alongside the original one-day target to test whether a longer horizon would improve signal-to-noise. Finally, we expanded the model comparison to three algorithms — logistic regression, random forest, and

XGBoost — all tuned under the same time-series cross-validation scheme. Table 1 reports the hold-out metrics for the 5-day target.

Table 1. *Model 3 hold-out results on the 5-day directional target. XGBoost achieves the highest accuracy, the only Model in the project to clearly beat the majority-class baseline.*

Model	CV AUC	Accuracy	AUC	Precision	Recall / F1
Majority class	n/a	0.5460	0.5000	n/a	n/a
Logistic Regression	0.5268	0.5375	0.5108	0.5486	0.8627 / 0.6707
Random Forest	0.5184	0.5525	0.5450	0.5569	0.8824 / 0.6829
XGBoost	0.5218	**0.5589**	0.5227	0.5579	0.9255 / 0.6962

The best configuration was the XGBoost classifier on the five-day target, which reached 55.89% directional accuracy on the hold-out set against a majority-class baseline of 54.60%. This was the only iteration in the project to clearly beat baseline. Figure 2 shows the confusion matrix and Figure 3 the ROC curve for this model.

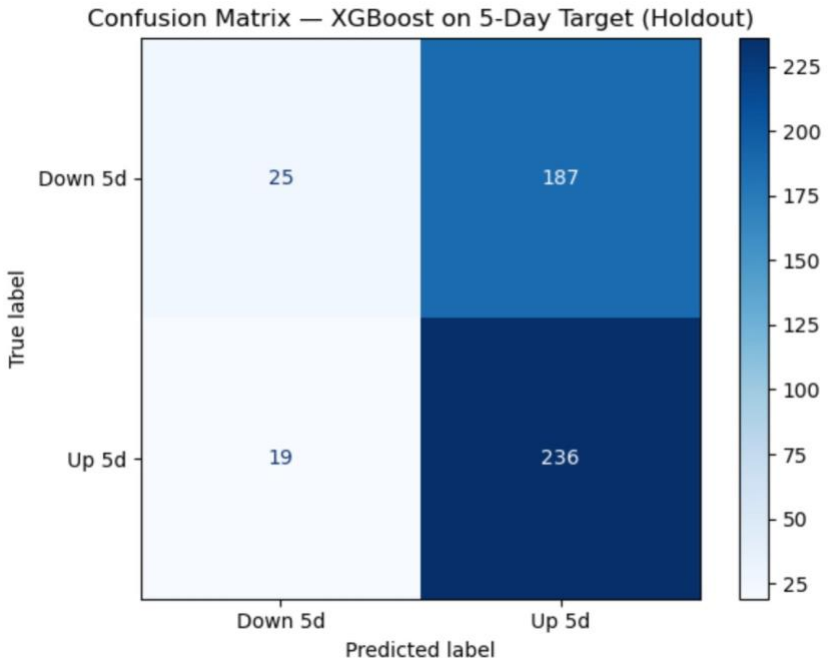

Figure 2. *Confusion matrix for Model 3 XGBoost classifier on the 5-day directional target. The model correctly predicts 236 of 255 actual up-days (92.5% recall) but only 25 of 212 actual down-days.*

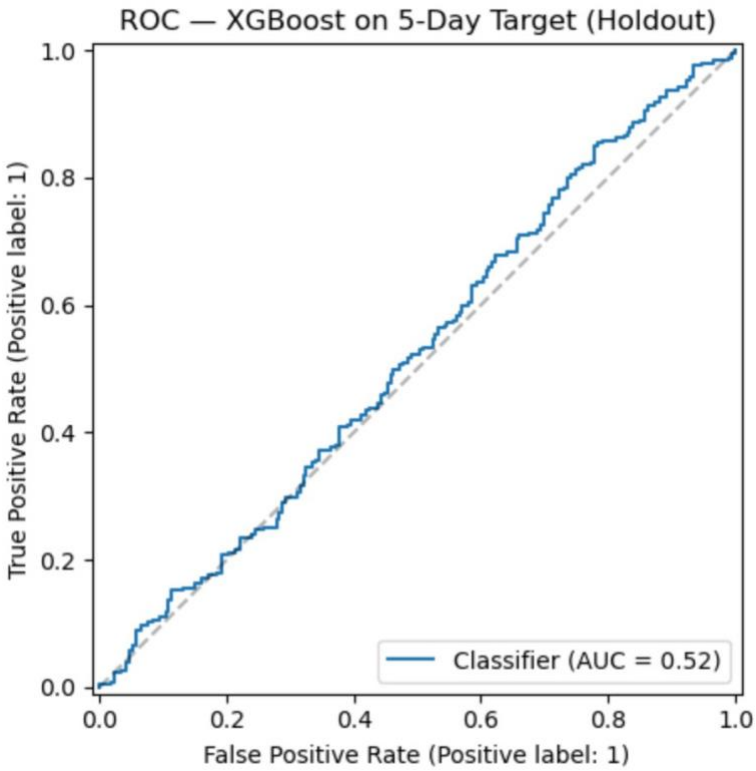

Figure 3. *ROC curve for Model 3 XGBoost classifier. AUC of 0.52 indicates weak but positive ranking ability.*

The confusion matrix reveals an important qualification to the headline accuracy number. The model predicts up on 92.5% of hold-out days; its advantage over the majority-class baseline comes more from class imbalance in the target (54.6% up-days) than from genuine separation between up-day and down-day predictions. This observation motivated the refinements in Model 4.

Model 4 — Regression, Threshold Tuning, SHAP, and Technical Indicators

Model 4 tried to improve on Model 3 in a few ways. We switched from classification to regression, predicting the actual 5-day return instead of just up or down. This gave us more information to work with since a predicted 0.1% return and a predicted 3% return are no longer treated the same. We added SHAP to better understand which features the model was relying on, and we brought in classic technical indicators: RSI, MACD, and the distance from the 52-week high. We also dropped the Nikkei feature because the Japanese market calendar didn't line up with US trading days and was costing us about 24% of our data.

To try to improve on Model 3's "always predict up" pattern, we tuned the decision threshold. Instead of using zero as the cutoff between up and down predictions, we tested different thresholds on the training data to find one that gave better accuracy.

Figure 4 shows the result of that threshold sweep. The accuracy actually gets worse as the threshold rises, which means the model is most accurate when it's predicting up on almost everything. This isn't really a tuned threshold in any meaningful sense. It's the model telling us

that its confident predictions aren't any more reliable than its uncertain ones, so the safest strategy is just to always predict up.

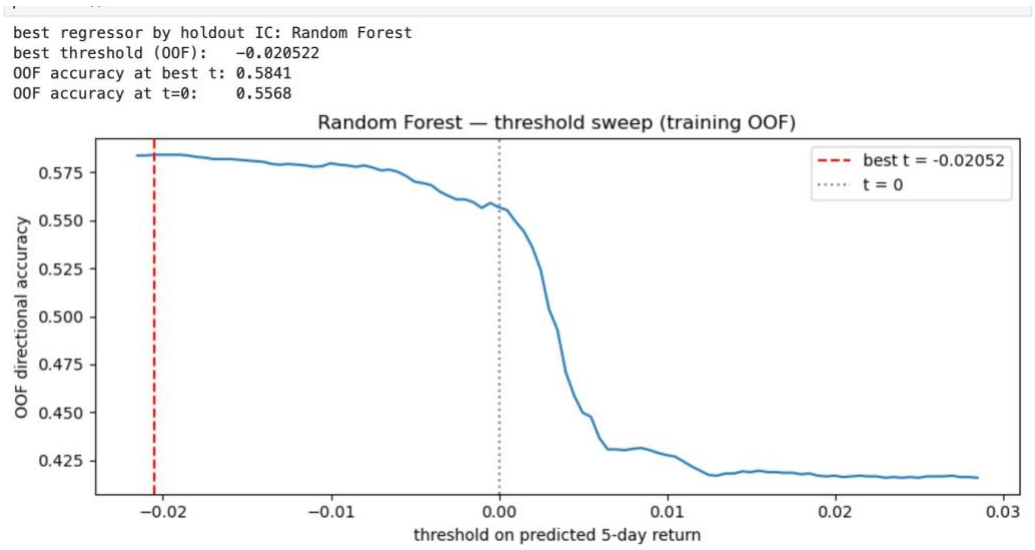

Figure 4. *Threshold sweep on training out-of-fold predictions for Model 4's best regressor (Random Forest). The monotonically declining accuracy curve indicates that the model's confident predictions do not reliably separate up-days from down-days.*

Table 2. *Model 4 hold-out directional accuracy across threshold choices.*

Configuration	Holdout Accuracy
Majority-class baseline	53.22%
Random Forest, threshold = 0	53.74%
Random Forest, tuned threshold	53.39%

The hold-out directional accuracy of Model 4's best configuration was 53.39% at the tuned threshold and 53.74% at threshold = 0, neither of which meaningfully exceeds the majority-class baseline of 53.22%. Figure 5 shows the confusion matrix at the tuned threshold. The model predicts up on 574 of 575 hold-out days, a degenerate pattern that is the direct consequence of the monotonic threshold sweep in Figure 4. This is not a flaw in the implementation but rather an honest reflection of the weakness of the underlying signal.

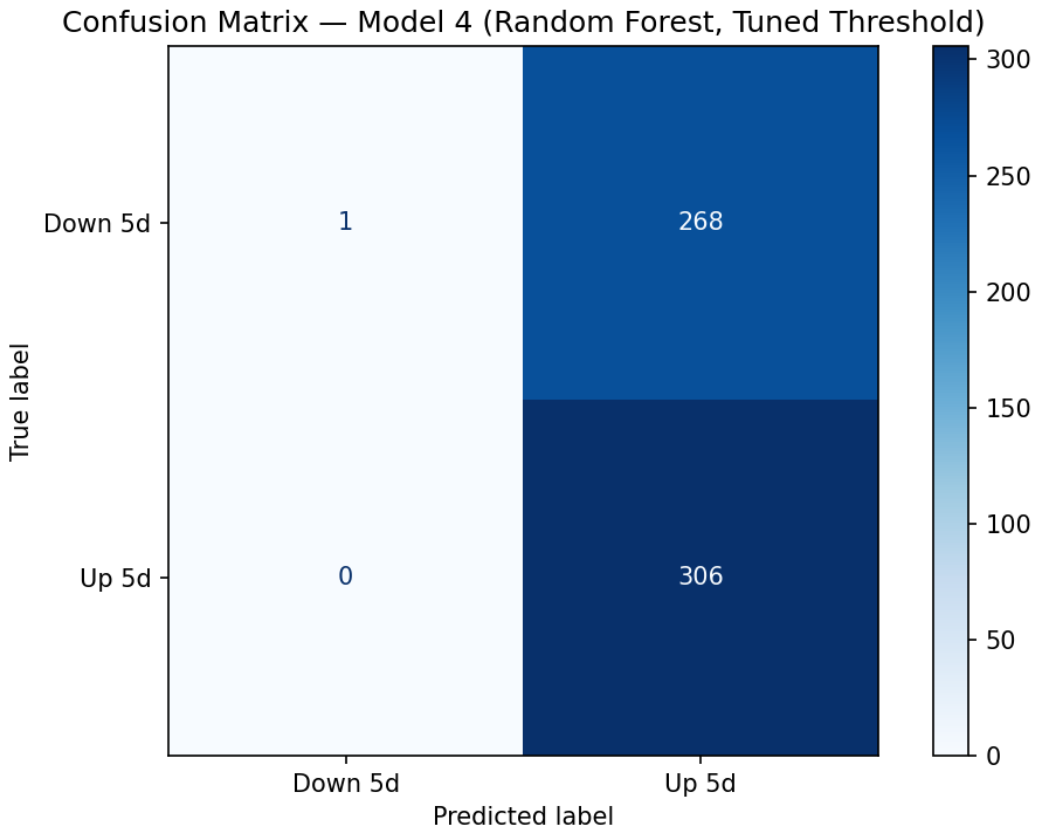

Figure 5. *Confusion matrix for Model 4 at the tuned threshold. The model predicts up on virtually every hold-out day.*

The SHAP analysis (Figure 6) ranks features by their mean absolute contribution to model predictions. The top fifteen features are primarily driven by return-based measures (RetEMA8, RetVol20, RetEMA4, VolRatioLag1), highlighting the importance of recent price dynamics. Two technical indicators- RSI_14 and MACD rank second and third overall, making them more influential than any cross-asset feature, including the supply-chain signal derived from TSM. This indicates that the model captures meaningful structure within AAPL's own return history, particularly at the longer horizons where these indicators are most effective. In contrast, cross-asset features generally rank lower, with only TSM and VIX level

appearing in the top ten.

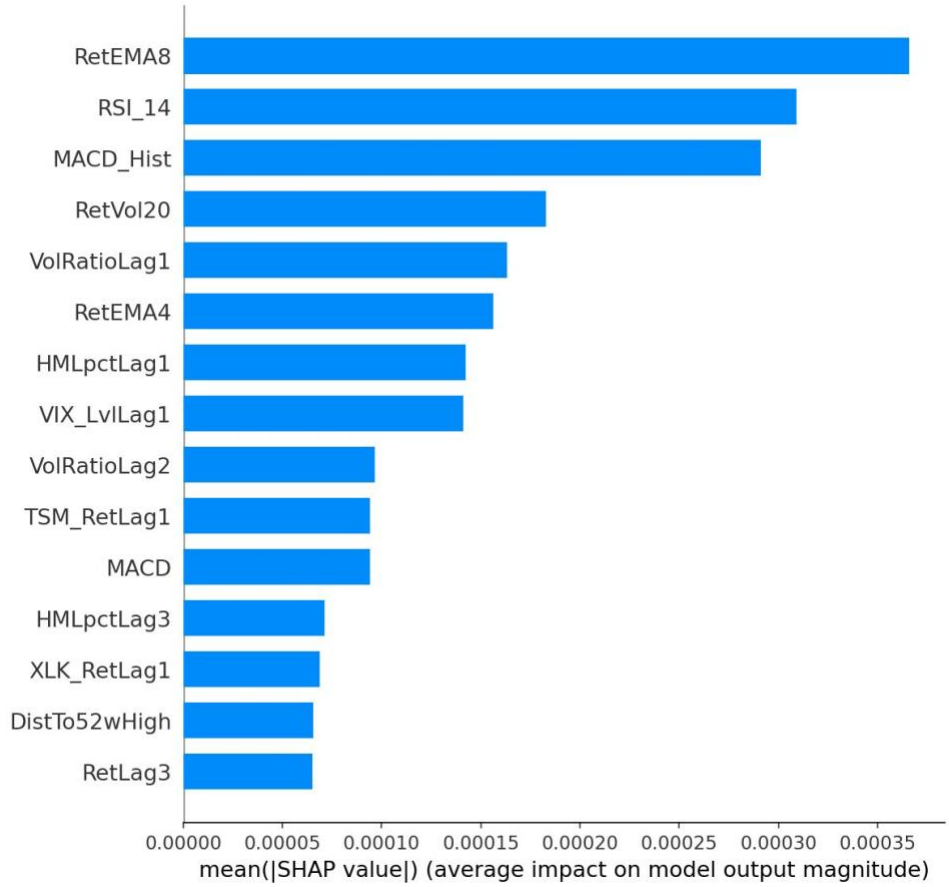

Figure 6. *Model 4 SHAP bar plot showing mean absolute SHAP values for the top 15 features. RSI_14 and MACD_Hist rank second and third, though absolute magnitudes are small across all features.*

Full SHAP Feature Ranking — Model 4

Table 3. *Complete SHAP feature ranking for Model 4, all 29 features. Absolute magnitudes are small across the entire feature set; the top feature shifts predictions by roughly four basis points on average.*

#	Feature	Mean |SHAP|	Group
1	RetEMA8	0.000366	Return
2	RSI_14	0.000309	Technical
3	MACD_Hist	0.000291	Technical
4	RetVol20	0.000183	Return
5	VolRatioLag1	0.000163	Return
6	RetEMA4	0.000156	Return
7	HMLpctLag1	0.000142	Return
8	VIX_LvlLag1	0.000141	Cross

#	Feature	Mean |SHAP|	Group
9	VolRatioLag2	0.000097	Return
10	TSM_RetLag1	0.000094	Cross
11	MACD	0.000094	Technical
12	HMLpctLag3	0.000071	Return
13	XLK_RetLag1	0.000069	Cross
14	DistTo52wHigh	0.000065	Technical
15	RetLag3	0.000065	Return
16	RetLag1	0.000059	Return
17	HMLpctLag2	0.000054	Return
18	QQQ_RetLag1	0.000046	Cross
19	SPY_RetLag1	0.000043	Cross
20	OMCpctLag3	0.000043	Return
21	OMCpctLag1	0.000043	Return
22	VIX_ChgLag1	0.000041	Cross
23	SOXX_RetLag1	0.000039	Cross
24	TNX_ChgLag1	0.000039	Cross
25	MACD_Signal	0.000037	Technical
26	RetEMA2	0.000035	Return
27	OMCpctLag2	0.000018	Return
28	RetLag2	0.000018	Return
29	VolRatioLag3	0.000012	Return

Conclusion and Next Steps

Conclusion

Across four iterative model builds, no configuration produced directional accuracy meaningfully above a majority-class baseline on out-of-sample AAPL return prediction at either the one-day or five-day horizon. This is a negative result, but it is a well-supported one. The progression from Model 1 (severely overfit due to non-stationary price features) through Model 2 (stationarity fixed, but no signal) through Model 3 (five-day horizon, richer cross-asset set, best numerical accuracy) to Model 4 (regression, threshold tuning, SHAP interpretability, technical indicators) represents a steady increase in methodological rigor.

Each iteration was motivated by a specific weakness in the prior one, and each delivered the intended improvement in diagnostic clarity. The final model is the most honestly evaluated one, and its lack of predictive edge is the correct answer to the research question given the features and horizon tested.

The project delivers value not by producing a tradeable model but by demonstrating a disciplined application of machine learning to financial return prediction: stationary feature construction, proper train/hold-out separation, time-series cross-validation with gaps, threshold tuning on out-of-fold predictions, and model interpretability via SHAP. These are the techniques that distinguish a defensible analysis from a data-mined one.

Next Steps

Further steps can be taken to improve the output of the model. First, we can assess if the model is fit to predict over larger periods of times; monthly, quarterly, and yearly predictions can be useful in the investing environment. Secondly further development of the model can be tried to see if by incorporating other metrics like some less obvious cross assets and indicators can drive higher performance. Some ideas include fundamental data (earnings surprise, revisions to consensus estimates, guidance changes), alternative data (shipping volumes, satellite imagery, web search trends), and unstructured data (news sentiment, earnings call transcripts). These data sources are harder to obtain but have the potential to carry signal that has not already been arbitraged away. Third, the evaluation could be extended with an economic backtest that converts predictions into a simple trading rule (go long on predicted up-days, flat otherwise) and compares cumulative return against buy-and-hold, with realistic transaction-cost assumptions. Even a statistically weak model can look different on a P&L basis, particularly in a trending market.

A fourth direction, worth noting more for its methodological interest than its expected return, would be to apply the same pipeline to a less intensively studied security. AAPL is an especially hard test case precisely because it is so widely followed. The same techniques applied to a smaller-capitalization stock, a less-developed commodity contract, or a cryptocurrency might produce different results. The framing would shift from "can we beat the market on the most efficient stock in the world" to "where in the cross-section of securities does weak-form efficiency break down," which is arguably a more answerable question at this scale.

AI Disclosure

Anthropic's Claude (Opus 4.7) was used throughout this project as a technical and editorial assistant. After models 1 and 2, Claude was consulted on ideas to improve model performance which we see the output with including more cross asset features, changing the evaluation method, and use of SHAP. On Saturday I attended Northwestern event on Healthcare Startups hosted by Professor Srinivasan which is where I got the idea to use SHAP. Claude was used to assist in formatting of final report to improve appearance.

All modeling decisions, final code execution, interpretation of results, and substantive writing were performed by the author. The choice of security (AAPL), the selection of cross-asset features, the decision to prioritize methodological rigor over optimistic result reporting, and the specific content of the conclusion and next steps reflect the author's own analysis and judgment as well.